Follow the Money Without Moving the Data

Privacy-limited questions between banks,
planned by an LLM agent, for anti-money-laundering

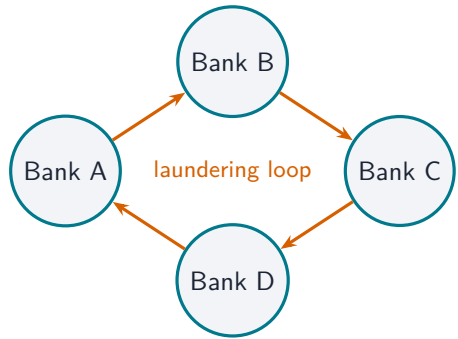
Team **Cache me if you can**: Abhishek · Bhavya · Shreyas Deb · Akshat Srivastava

LMA Group Project 2026 · `github.com/BhavyaV05/Transaction-Thieves`

The problem: every bank sees only its own slice

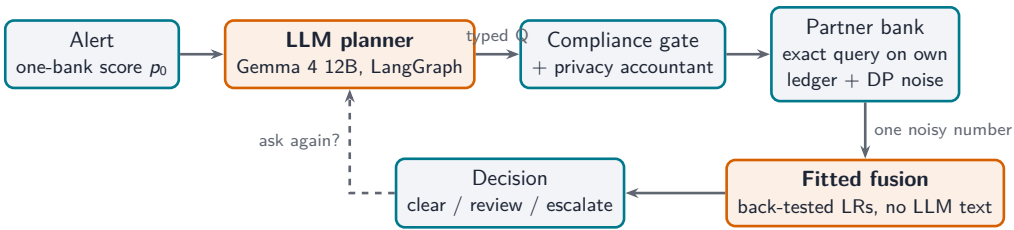
- Launderers route money **across banks** on purpose.
- A model trained on one bank's view misses most of the pattern.
- Banks **cannot pool data**. Legal sharing (US 314(b), MAS COSMIC) is slow, manual, and sends raw records.

Our question: can a bank gain by asking other banks a few small, *private* questions, without any record leaving them?



Bank A sees only 2 of 4 hops

The idea in one picture



- **No transaction record leaves a partner bank**: only one number with differential privacy noise.
- Budget $\varepsilon = 2$ per alert, cap $\varepsilon = 8$ per partner account. LLMs only see aliased accounts.

Three design rules that keep the LLM honest

1

Back-test before trust

Every question, even one the LLM just wrote, must prove on past alerts that its answer is informative (likelihood ratio + e-value FDR control).

2

A formula decides

The LLM plans. The final probability comes from $\text{logit}\, p_0 + \sum w \log \mathrm{LR}$, numbers only. Prior work: an LLM investigator trusted its own reasoning and did *worse* than its base model.

3

One gate for everyone

Our planner and every external agent use the same gate, accountant, simulated banks, backbone and call limits. Only the control loop differs.

Data: IBM AMLworld, turned into 30 banks

- **HI-Small** (5M transactions) for development, **HI-Medium** (32M) for the headline.
- We built **30 bank views** with private ledgers, reply latency, declines and untrusted text notes.
- **Time-respecting split**: days 1–9 / 10–12 / 13–16. Test split locked in code until the frozen final run.
- **Conformal intake** passes only alerts the bank's model cannot settle.

HI-Medium test	count
Alerts	4,478
headline period	3,873
laundering	975
Askable (a partner can answer)	**771**
Random subset, askable	208

What we built and ran

Detection baselines (reproduced)

- GFP + LightGBM: F1 62.95 vs 62.86 published
- Multi-GNN (official code): F1 67.64 vs 68.16
- MEGA-GNN (partial), FraudGT (500 epochs)
- DP-FedAvg (Flower + Opacus), 6 ε levels $\times$ 3 seeds
- Layer-wise exchange (Ceydeli et al.), 8 arms

Planners compared

- Random, fixed expert order, EIG, learned policy, enumerator of every question, raw record sharing

13 LLM agents ported into one harness

- AML: Sharma, Torres, Co-Investigator, FinFRE-RAG
- Info-seeking: BED-LLM, Uncertainty of Thoughts, debate
- General: ReAct, Reflexion, ReWOO, LATS, LLMCompiler, DSPy-tuned ReAct

Our LLM side

- Backbone sweep of 7 open models (E18)
- QLoRA student planner, 3 seeds (E16)
- Qwen3-4B fast model with conformal cascade + injection screen

Compute: days of GPU time, one frozen test run

44
GPU-hours
final test run alone (RTX 5070 Ti)

∼60
GPU-hours
validation reruns on the LangGraph pipeline

33
systems
run on validation fit / cal / eval

12k
verdict rows
in the final run (GPU + CPU)

- Detection models trained for days on the 5070 Ti and a 3070 Ti laptop; RAG and injection benchmarks on the *ada* HPC cluster.
- Hypotheses H1–H10, thresholds and Holm order **pre-registered and frozen with a SHA-256 hash** before the test split was opened.

How we score: gap closed

$$\text{gap closed} \;=\; \frac{B(\text{one bank}) - B(\text{system})}{B(\text{one bank}) - B(\text{see everything})}$$

- B = Brier score (lower is better)
- $\mathbf{0}$ = no better than the bank alone
- $\mathbf{1}$ = as good as pooling every bank's data
- 95% cluster-bootstrap intervals, Holm correction

Detection model (3,873 alerts)	Brier
GFP one bank	0.1585
GFP see everything	0.0854

A big gap to close: precision in the top 10% goes from 0.72 to 0.98.

Headline: questions do not beat the bank alone

	System (771 askable test alerts)	Brier	Gap closed [95% CI]
Reference	One bank only	0.1668	0
	See everything (GFP)	0.0966	1
	Raw records from 3 banks, no privacy	0.1640	0.040 [−0.107, 0.161]
Non-LLM	Fixed expert order	0.1694	−0.036
	Enumerator	0.1708	−0.057
	Expected information gain	0.1823	−0.220
External	BED-LLM (best on validation)	0.1956	−0.410
Ours	**Compose,** $\varepsilon = 2$	**0.1673**	**−0.007** [−0.038, 0.026]

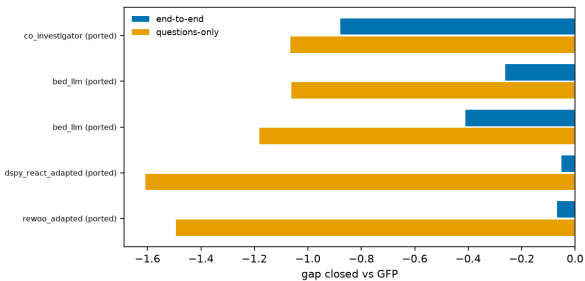
- Even **raw, noise-free** records close only 4%: the limit is the data, not the privacy noise.

Pre-registered hypotheses on test

	Hypothesis	Estimate [95% CI]	p_{Holm}	Verdict
H1	Gap closed vs one bank	−0.007 [−0.038, 0.026]	1.00	not supported
H2	Benefit vs raw sharing	−0.165 [−2.88, 0.75]	1.00	not supported
H3	LLM beats simple planners	−0.001 [−0.006, 0.003]	0.33	not supported
H4	Back-testing helps	−0.005 [−0.010, −0.000]	0.072	not supported
H9	**Beats best external agent**	**0.266** [−0.022, 0.799]	**0.004**	**supported**
H7	Formula vs LLM deciding	−0.000 [−0.004, 0.004]	1.00	not supported

- H4 came close but its false-discovery audit failed. H5, H6, H8, H10 were not run in the reduced final plan.

Where we win: against published LLM agents



Agent (208 subset)	Brier
Ours	**0.1576**
DSPy ReAct (adapted)	0.1623
ReWOO (adapted)	0.1638
BED-LLM	0.1805
Co-Investigator	0.2334

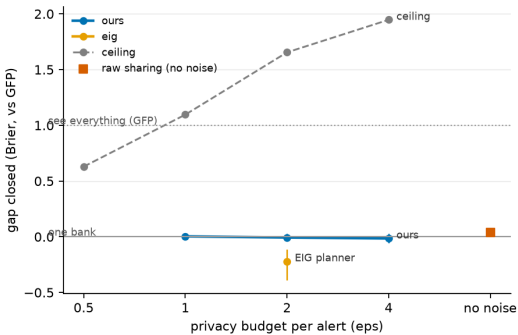
- Agents that decide for themselves drift away from a well-calibrated p_0.
- Co-Investigator gives no verdict on 71% of alerts.
- BED-LLM uses 3× our LLM calls.

Ablations: what earns its place

Switch off (208 askable alerts)	Brier	Δ vs full [95% CI]	
Full system	0.1576	–	
LLM may only pick from a fixed list	0.1656	+0.0080 [0.0031, 0.0131]	**composing helps**
No back-testing	0.1626	+0.0049 [0.0003, 0.0099]	**back-testing helps**
No stated hypothesis (act only)	0.1592	+0.0016 [−0.0026, 0.0059]	not significant
LLM decides stop / escalate	0.1577	+0.0001 [−0.0035, 0.0038]	no difference

- Without back-testing the agent asks **twice as many questions** and costs twice the GPU time, for a worse score.
- From validation: naive fusion ($w = 1$) moved scores the wrong way (double counting p_0), so we fitted per-family weights.

Privacy and cost



- Flat curve: $\varepsilon = 1, 2, 4$ give Brier 0.1666, 0.1673, 0.1679.
- Spends $\varepsilon \approx 1.46$ of 2 per alert.
- Per alert: **2.4 LLM calls**, 12.9k tokens, 29 GPU-seconds.
- On all 3,873 alerts we **do no harm**: $+0.0001$ Brier vs one bank (BED-LLM $+0.0057$).

The LLM side: backbone, fine-tuning, safety

Backbone sweep (E18)

7 models, 2 stages. **Gemma 4 12B** won (0.1741 vs 0.1844 Qwen3.8-27B). "Thinking" mode spent its whole budget reasoning and was dropped.

QLoRA student (E16)

Qwen3-8B, LoRA r16, 3 seeds. Learned the format (token acc. 0.94) but planned worse than the 12B teacher (0.223 vs 0.201).

Fast model + safety

Qwen3-4B with conformal cascade: every error bound held. As an injection screen it cut ReAct's attack success from 5–8% to **0%**.

Limitations and recorded deviations

Limitations

- Synthetic data and simulated banks
- Only $\sim$20% of alerts are askable
- Privacy per transaction, not per customer
- Reduced final run: some arms not run

Deviations (all documented)

- GFP see-everything rerun did not exactly reproduce; scores archived for a rerun
- GNN official loaders leak future edges, so GFP is the only upper limit
- CPU arms rerun with the frozen p_0; FraudGT scored before the freeze

Take-aways

1. **Honest negative result.** Private questions between banks do not measurably beat a good one-bank model on AMLworld, and neither do raw records. The bottleneck is what a partner can add, not privacy.
2. **The safeguards work.** Back-testing plus a formula never hurts the base model, and it significantly beats published agents that let the LLM decide (H9).
3. **Inside the system**, composing new questions and back-testing them both measurably help.
4. **Reproductions** of GFP, Multi-GNN, MEGA-GNN and FraudGT on a time-respecting split, plus a finding that the GNN loaders leak future edges.

Code, report and a README with full reproduction steps:
`github.com/BhavyaV05/Transaction-Thieves`

Thank you

Questions?